

Designing Embodied Metaphors and Analogies for AI Literacy: An Iterative Study with Experts, Teachers, and Children

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Teaching complex AI concepts to children presents significant challenges, as educators often struggle to both grasp nuanced mechanisms and translate them into engaging learning experiences. Drawing on embodied cognition and analogical reasoning theories, this work aims to create embodied metaphors and analogies that simplify the mechanisms and ethical issues of AI recommendation systems for middle-school students. We employed a multi-phase design pipeline grounded in the Design-Based Research (DBR) methodology to integrate feedback from key stakeholders, including domain experts, students, and teachers, for linking abstract AI concepts to concrete and familiar experiences. This process began with collaborative brainstorming with seven AI experts to generate diverse embodied representations. The resulting designs underwent mixed-method evaluations with both middle-school students and teachers, which revealed positive shifts in student attitudes and provided insights for further iteration. The design cycles ultimately uncovered key challenges, such as balancing feature fidelity with cognitive overload, and yielded design implications for developing effective embodied and analogical representations for children's AI literacy.

CCS Concepts: • **Applied computing** → **Interactive learning environments**; • **Human-centered computing** → **Empirical studies in HCI**; *User centered design*;

Additional Key Words and Phrases: Embodied learning, Analogical learning, AI literacy, AR application, Design-based research, Iterative design process

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1 Introduction

AI recommendations shape our daily decisions, with no exception for our young generation [36]. The convenience of navigating personalized content goes hand-in-hand with the notorious “filter bubble” effect [56], which may decrease people’s exposure to diverse options and opinions. Children are especially vulnerable to this due to their underdeveloped critical thinking skills [44]. To foster AI literacy in young learners and empower them to address ethical concerns [59], analogy-based learning [28] and embodied learning [62] offer promising approaches to making abstract mathematical and computational concepts more accessible [18].

Drawing on theories of embodied cognition [40] and analogical reasoning [26], embodied metaphors and analogies can lower children’s cognitive load for understanding abstract, unfamiliar domains by linking them to concrete, familiar experiences [22]. Embodied representations convey abstract concepts through physical or sensory experiences, while analogies clarify them by relating these concepts to familiar ideas; both approaches make abstract ideas more intuitive. Effective design of embodied and analogical representations hinges on establishing structural connections between the target concept’s sub-components and corresponding elements of the source domain, a challenge for both teachers and AI experts. Teachers often struggle to grasp nuanced AI concepts [53], while AI experts must translate these complex ideas into representations that resonate with children.

Our work proposes an iterative design pipeline to develop a set of embodied metaphors and analogies for key concepts around AI recommendation systems. The representations are implemented in a tablet-based **Augmented Reality (AR)** application that aims to teach AI literacy to middle-school students. This iterative design pipeline consists of collecting design ideas by brainstorming with AI experts, iteratively evaluating the AR prototype with students, and conducting focus group interviews with teachers. First, we identified several themes for making AI concepts engaging and accessible to children with embodied analogies. Based on these insights, we developed an AR application that used bees, flowers, and gardens as embodied analogies for complex AI mechanisms. Mixed-method evaluations with 14 middle-school students and 4 STEM teachers indicate the design’s effectiveness and future iterations needed.

We aim to address the research question: How can an iterative, multi-phase design pipeline produce and refine embodied analogies for teaching AI literacy? We specifically investigated three sub-RQs:

- RQ1. Ideation:* What design concepts emerge from brainstorming with AI experts?
- RQ2. Formative evaluation:* What learning challenges and affordances do middle-school students reveal when using the AR prototype?
- RQ3. Refinement:* How does teacher feedback from focus-group interviews inform the next design iteration?

Our findings highlight the challenges in fully conveying algorithmic details through embodied and analogical learning, cognitive overload from the numeric representation of data, and an overemphasis on certain visual features in AR. We further derived iterative refinements, such as integrating different representation modalities and illustrating the consequences of filter bubbles more explicitly. This work makes three main contributions:

- (1) It introduces an iterative design pipeline that integrates insights from AI experts, middle-school students, and teachers to develop embodied and analogical representations for key AI concepts.
- (2) It provides mixed-method evaluation results demonstrating effectiveness for learning while identifying critical design challenges.
- (3) It summarizes a set of design implications for embodied and analogical representations for developing children's AI literacy.

2 Related Work

2.1 Embodied Metaphor for AI Literacy

Embodied cognition is a cognitive theory that suggests our thinking processes are deeply connected to our physical actions [61]. This idea contrasts with traditional cognitive science, which views the mind as a computer that processes symbolic inputs into outputs [60]. Embodied cognition underscores the interconnectedness of our senses, perceptions, and actions. This concept changes how we see the relationship between the body and the brain, viewing them as partners in thinking processes [14]. The body helps determine what sensory information is necessary for a perceptual experience and structures that information, working in tandem with the brain to produce cognition [60]. Embodied learning translates these theoretical insights into educational practice, proposing that learning occurs when bodily movements, physical interaction, and sensorimotor abilities are linked with the learning content [5, 6]. It operationalizes the principles of embodied cognition through learning experiences that actively engage the body in the learning process, such as gestures, physical manipulation of tangible objects, or movement-based interactions [46].

Johnson-Glenberg et al. proposed a taxonomy for embodied learning, identifying three primary factors: motoric engagement, gestural congruency, and perceived immersion. Motoric engagement involves full-body movement through locomotion. Gestural congruency relates to how well gestures align with the learning content. The gestures should be mapped to the content to reify the learning construct. The third factor, perceived immersion applies specifically to the experience in virtual reality and other similar technological tools. It depends on the type and configuration of the content's display. These factors combine to form four distinct degrees of embodiment [37]. The first degree has the learner seated with minimal upper body movement, no gestural relevancy, and non-immersion. The second degree sees the learner generally seated with upper body movement, interactive interfaces but limited gestural relevancy, and a less immersive environment. The third degree includes whole-body engagement without sustained locomotion, some gestural relevancy, and high immersion. The fourth degree involves locomotion with high sensorimotor engagement, consistently mapped gestures, and a highly immersive environment.

Embodied learning has demonstrated its effectiveness in areas such as computational thinking and math education [9, 52]. Comparative research indicates that students in embodied learning environments often achieve greater learning gains compared to those in traditional settings [20, 29, 73]. These studies suggest that embodied interactions support learning by grounding abstract concepts in concrete representations [12, 33], reducing cognitive load [10, 20, 22], and fostering creativity, self-exploration, and self-regulation [33, 35, 64].

Building on these affordances, emerging research explores how embodied interactions can deepen understanding of AI systems. For example, Ghajargar et al. proposed "Graspable AI", a design framework that makes AI tangible by integrating insights from Explainable AI, Data Physicalization, and Tangible User Interfaces [30, 31]. Preliminary evidence suggests that embodied interactions can improve user awareness and control over AI systems, enhance understanding of AI concepts, and increase motivation and engagement in learning AI [4, 16, 39, 42, 50]. However, establishing

clear structural connections between the sub-components of AI and their physical counterparts is inherently complex. Another challenge stems from the need to balance complexity and simplicity so that the embodied metaphors can be intuitive enough for children to grasp while maintaining the depth required to convey important AI literacy. Therefore, both pedagogical knowledge and AI expertise are required for designing concrete embodiments to capture the essence of AI concepts, such as algorithms, data processing, and ethical considerations.

2.2 Analogy Design for AI Literacy

Analogy-based learning is a cognitive process where people make connections between seemingly unrelated ideas or concepts for learning [28]. Stages in this reasoning process involve (1) retrieval, which occurs when a person is reminded of a prior, relationally similar case while thinking about a current topic; (2) mapping, which involves comparing two cases that are simultaneously present; (3) evaluation, where the analogy is judged, along with any inferences that have been generated. Three factors define a good analogy [27, 57]. First, systematicity refers to whether two analogs share a deeply connected relational structure. Second, transparency suggests how similar two domains appear to be. Third, literal similarity matches as domains that appear to be similar superficially are more likely to be understood than pure relational analogies.

Using analogies for teaching and learning can be double-edged. Historically, criticisms in computing education highlight that analogies might oversimplify complex concepts or introduce irrelevant and potentially misleading features [8]. For instance, Dijkstra et al. argued that computing was too novel to be effectively explained by analogies. Halasz and Moran reinforced concerns that analogies introduce misconceptions through superficial mappings rather than structural relationships. Recent arguments against analogy use in computing education center around several core concerns. The first is the limitation of analogies: specifically that learners may apply analogies beyond their valid contexts. This over-extension can lead to confusion and incorrect inferences, particularly when analogies are used across different domains. Second, analogies rely heavily on learner relevance. Because learners' prior experiences vary, an analogy that resonates with one group may fail for another if the source domain is unfamiliar or culturally irrelevant [8]. Structural soundness is another concern. Because analogies frequently arise spontaneously, they may not always clearly and accurately represent the intended structural relationships. Addressing these issues requires deliberate analogy design and careful instructional guidance [8].

Considering the potential drawbacks of analogy use in computing education, existing research has drawn on Zone of Proximal Development theory [23] to propose a human-AI analogy framework for AI literacy development in upper primary classrooms [17]. This framework leverages the human-AI comparisons as a scaffolding tool to clarify AI concepts and processes. By using familiar human characteristics as analogies for AI functionalities and emphasizing both the similarities and differences between them, it creates meaningful learning opportunities that promote a deeper understanding of AI among students [18].

Designing effective analogies for representing AI concepts to develop children's AI literacy presents several challenges. One major hurdle is translating the abstract and multifaceted nature of AI—encompassing algorithmic processes, data flows, and ethical considerations, into relatable experiences without oversimplifying critical elements [18, 28]. Educators often face difficulties in fully grasping nuanced AI concepts [53], which can impede their ability to identify and communicate the core ideas effectively. Simultaneously, AI experts may struggle to pinpoint engaging and age-appropriate source domains that resonate with young learners. Balancing technical accuracy with pedagogical accessibility, while avoiding cognitive overload, requires a carefully calibrated, iterative design process that actively involves AI experts, teachers, and students.

2.3 Teaching Recommendation Systems in K-12

Emerging research has been done over the past few years to investigate ways to promote AI literacy among audiences without a technical background. AI applications and AI ethics are the emerging topics among the current frameworks, standards, and curricula for AI for K12 [2, 43, 48, 66]. The following are identified as important components: acknowledging the mechanism behind AI applications, recognizing different stakeholders and end-users of AI systems [2, 43], and understanding both the negative and positive impacts of AI applications on society [48, 66]. The AI recommendation system is one of the AI systems that relates most closely to the young generation's daily lives. The applications of recommendation systems, such as e-shopping, e-learning, and e-resource services, are used frequently by adolescents [51]. The information provided by the recommendation systems in these applications is not always appropriate or beneficial for children and adolescents [3]. Adolescents will benefit if they acknowledge how the recommendation systems work and their related ethical issues when they interact with these applications and consume the information.

Despite the importance of teaching recommendation systems to K12 children, there are scarce curricula and technologies to support learning and teaching recommendation systems. MIT Media Lab held AI summer workshops that had one unit to redesign YouTube's recommendation system by identifying stakeholders [2, 43]. Researchers have also explored informal learning experiences in museum exhibitions to foster youth's critical thinking skills around AI [44]. This work explores the design space of embodied metaphors and analogies for representing AI concepts through an iterative design and evaluation process with AI experts, teachers, and students. We hypothesize that collaboration with AI experts will yield an initial set of embodied metaphors and analogies aligned with key elements of AI literacy. Subsequent evaluations with teachers and students are expected to provide further insights to refine these designs and enhance student engagement and learning.

3 Iterative Design Process Overview

For designing BeeTrap, we employ design research as our primary methodological approach. This approach yields artifacts and systems as its major outcomes [25]. We aim to follow the criteria for design research in HCI. This includes maintaining a thoroughly documented process, ensuring that the design innovation is grounded in and builds upon existing literature, being contextually relevant, and laying the groundwork for future research [74]. The design thinking process is applied to promote an iterative cycle of design and evaluation [45].

In the first study, we conducted a participatory brainstorming session with AI and HCI experts to explore the use of embodiment and analogy for a child-centered AI learning experience (Section 4). AI experts were chosen over K-12 teachers in this phase due to K-12 teachers' limited AI knowledge, making it difficult for them to identify or solve design challenges. Seven AI experts proposed design ideas of using physical objects, movements, role-playing, and storytelling to make abstract AI concepts, such as data representations, training processes, and ethical concerns, more engaging and accessible to children. However, designing tangible interactions that are effective for children's learning remains challenging. Based on these insights, a set of design elements was developed to bridge abstract AI ideas with everyday experiences, leading to an AR application that uses familiar natural metaphors like bees, flowers, and gardens to enhance learning through playful interaction (Section 5). This design outcome needs to be iteratively evaluated by teachers and students to identify more learner-centered challenges and design insights.

In the second study (Section 6), we examined whether using physical distance as an embodied representation, along with analogies (e.g., a garden with flowers and bees), could effectively teach young learners about filter bubbles and AI recommendation systems. First, we conducted a proof-of-concept evaluation study with middle-school students, which indicates the effectiveness of

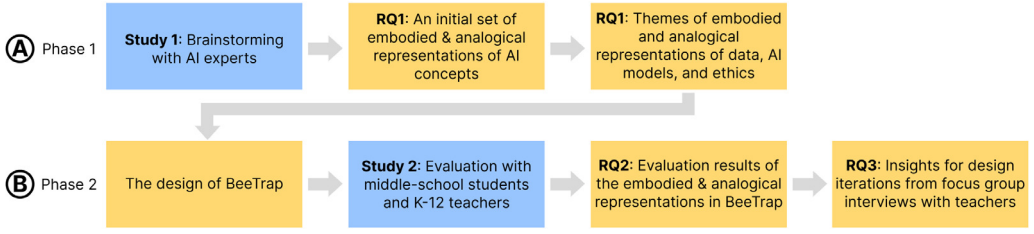


Fig. 1. An overview of the iterative design process of embodied and analogical representations in BeeTrap for child-centered AI learning experiences: (A) Study 1 with AI experts (Section 4) that lead to the design of BeeTrap (Section 5); (B) Study 2 evaluating BeeTrap with K-12 students and teachers (Section 6) to analyze and identify design iterations needed for BeeTrap (Section 6.2).

BeeTrap in teaching these concepts (Section 6.2.1) while also identifying learning challenges. Second, feedback from a focus group interview with K-12 teachers provided insights for design iteration. Our findings suggest how embodied metaphors and analogies can be designed and iteratively refined to effectively represent key AI concepts for children’s AI literacy (Section 6.2).

Figure 1 provides an overview of the iterative design process.

4 Study 1: Brainstorming with AI Experts

4.1 Research Method

4.1.1 Participants. We invited seven AI researchers for individual expert brainstorming interviews. All the experts (six Ph.D. level, two Ph.D. students) have teaching experience in AI and/or HCI.

4.1.2 Study Procedure and Data Collection. The seven semi-structured interviews each lasted approximately 2 hours and were conducted by two investigators. The goal was to collect specific ideas regarding using AR and/or tangible interfaces for teaching AI-related knowledge to K-12 students. As the guidance and impulses for the discussion, we provided major AI knowledge components under four AI literacy categories [39, 48], including (1) data representation, (2) AI model, (3) AI applications, and (4) AI ethical issues. In addition, we also introduced four unique design patterns afforded by AR for education [41]: (1) 3D space visualization; (2) present learning content by spatially and/or temporally integrating the physical and virtual world; (3) provide richer digital information to guide learners’ attention; (4) afford natural learner interaction and enable embodied learning.

The interviews comprised elements of brainstorming and 3D prototyping. Besides pen, paper, and some 3D prototyping materials, we printed the design template for experts to quickly demonstrate and sketch their ideas of using AR or tangible interfaces to teach AI knowledge. We recorded the audio for each session for documentation and took notes. We also collected all the design artifacts from each session. Three authors discussed, labeled, and grouped these notes.

4.1.3 Measure and Data Analysis. After cleaning up the incomplete ideas (i.e., the ideas that don’t really connect with using AR to teach AI to young learners), we collected 47 ideas in total from the seven experts who participated in our brainstorming interviews. In the last two brainstorming interviews with the experts, we observed data saturation as all the ideas generated in those sessions shared the same core designs as the ones generated in the previous sessions.

4.2 RQ1 Findings: Themes of Embodiments for AI Concept Learning

Through initial design discussions with seven AI experts, we collected 47 ideas for the 4 AI literacy categories (see examples in Figure 2). The key concepts span from multidimensional data and kernel

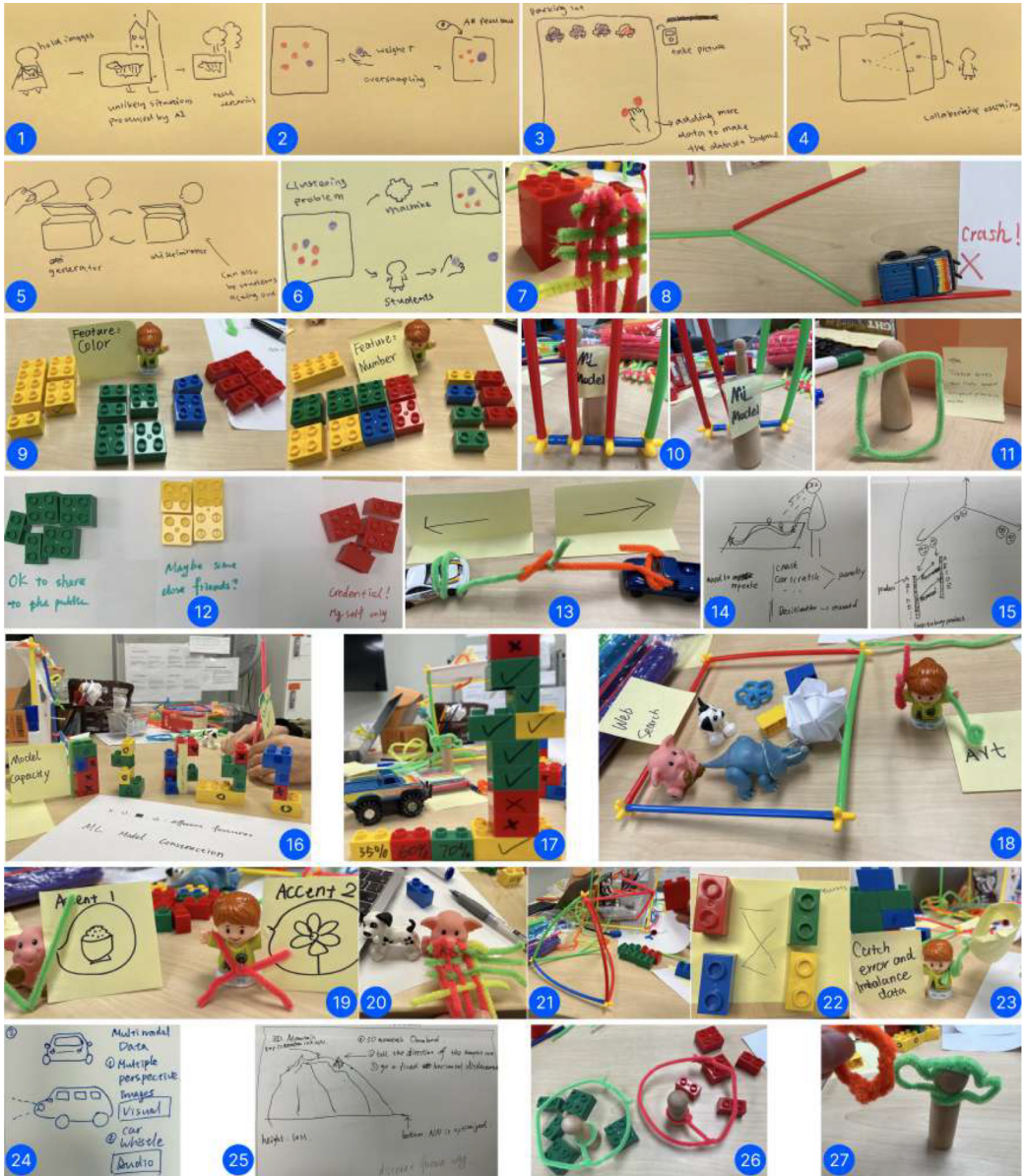


Fig. 2. Ideas collected from the brainstorming sessions with AI experts (Study 1): embodied representations for learning AI concepts around data representations (ideas 2, 9, 15, 17, 19, 20, 24, 26), AI algorithms (ideas 2, 4-8, 10, 13, 14, 16, 18, 21, 22, 25, 27), and AI ethical issues (ideas 1, 3, 8, 11, 12, 23). See more detailed descriptions in Section 4.2.

to classification models and data bias. We identified three themes in embodiments for making AI concepts engaging and accessible to children using embodied metaphors and analogies, along with challenges in their implementation.

Embodied Representations for AI Concepts around Data Representations. Experts have developed embodied approaches to illustrate the data representation, input, and output in AI algorithms using

physical objects and movements. Two distinct types of data representations have been identified in these processes. In the first approach, when the learning experience emphasizes multiple features of the data, various physical elements are used to express a multidimensional (e.g., Figure 2.9) and/or multi-modal (e.g., Figure 2.19) feature space. In the second approach, individual data samples are embodied as dots, highlighting discrete data points (e.g., Figure 2.2). The physical space or distance is generally used to denote the distance between data points (e.g., Figure 2.26). Although abstract concepts such as data distance, similarity, and modality can be explored through hands-on interactions, designing tangible representations that effectively convey these ideas remains challenging.

Embodied Representations for AI Algorithms. Experts identified AI training processes as one of the fundamental mechanisms that can be demonstrated through physical objects or narrative techniques such as storytelling. They proposed tangible representations as embodiments for concepts such as gradient descent, feature selection, and model updates. For example, some ideas use the weights of physical objects to embody the abstract concept of weights of features in AI algorithms (e.g., Figure 2.2); some use embodied representations to compare different models, such as paths for decision trees (e.g., Figure 2.8), boxes for generative AI (e.g., Figure 2.5), knitted fabric for matrix factorization (e.g., Figure 2.7), building construction (e.g., Figure 2.16), geometric relations (e.g., Figure 2.21), and hiking trails (e.g., Figure 2.25); Some ideas create analogies, such as vehicles (e.g., Figure 2.13), traffic management (e.g., Figure 2.14), and animals (e.g., Figure 2.18). The main challenge is simplifying these complex models without sacrificing their critical details.

Embodied Representations for AI Ethical Issues. Abstract ethical considerations of AI can be embodied through analogies, role-playing, and storytelling to emotionally engage children and improve retention. For example, role-playing scenarios, such as self-driving cars (e.g., Figure 2.8), corner case evaluation (e.g., Figure 2.1), data privacy (e.g., Figure 2.12), or problem-solving tasks in real-world (e.g., Figure 2.3) or imaginative (e.g., Figure 2.23) contexts, can stimulate discussions about ethical dilemmas around AI technologies; physicalizing or visualizing the ethical issues caused by AI applications also makes abstract concepts more concrete (e.g., Figure 2.11). However, crafting compelling stories that effectively convey complex issues like fairness and bias, while maintaining structural similarity to the AI concepts, is a significant challenge.

Summary. First, for data representation, experts proposed using physical objects and movements to illustrate multidimensional feature spaces and discrete data points, despite challenges in conveying abstract notions like data distance and similarity; second, for AI algorithms, tangible embodiments (e.g., weights, paths, vehicles) and analogies were suggested to represent training processes and model comparisons, with the difficulty lying in simplifying complex models without oversimplification; and third, for AI ethical issues, approaches such as role-playing, storytelling, and 3D visualization were recommended to foster engagement with ethical dilemmas, although crafting narratives that maintain the structural integrity of complex concepts like fairness and bias remains challenging.

Through further analysis and synthesis of ideas generated from expert brainstorming, we designed a list of embodiments of AI concepts that are compatible with each other and bridge the gaps between abstract AI concepts and concrete bodily experiences:

- (1) Multidimensional data points are embodied with physical features of objects; the data similarity/distance is embodied with physical distance.
- (2) Feature engineering is embodied with physical toolkits and real-time visual feedback, along with similarity-based models embodied by the physical space.
- (3) AI ethical issues are embodied through 3D visualization and full-body movements and role-play to address the perspective-taking of different stakeholders involved in AI technologies.

To create a specific design case for further evaluation and iteration with learners, we contextualized the embodiments above in a bee pollination context, which provides bees, flowers, and the garden as analogies for key concepts related to AI recommendation systems. The familiar context of nature allowed us to transfer children's existing knowledge of biology to complex AI mechanisms. Both visual and physical activities in this learning application solidify abstract AI concepts through a playful experience involving hands-on exploration and experimentation. The detailed design of BeeTrap is described in Section 5.

5 A Design Case: BeeTrap

5.1 Learning Objectives

Based on the major categories of AI literacy [48, 55], we defined the learning objectives centered around AI recommendation systems and the filter bubble ethical issue:

- (1) *Data*: (a) Apply multidimensional data objects to describe daily objects; (b) compare similarities and dissimilarities between multidimensional data objects.
- (2) *AI Model*: (a) Understand the concept of a similarity-based ML model; (b) apply feature selection to achieve better model performance.
- (3) *AI Application*: Understand basic concepts involved in a content-based recommendation system.
- (4) *Ethical Issues*: Understand the definition of the filter bubble effect and its impact on a person's daily life.

5.2 The Initial Set of Analogical and Embodied Representations

Our design is mainly informed by two learning science theories. First, analogical reasoning supports learners in developing new knowledge through well-designed relational connections with familiar domains [15, 27, 58]. Such re-representation can shift learners' perspectives and support active learning for the underlying relational mapping between the source domain and the target domain [28]. Second, embodied interaction promotes active learning through engaging and accessible manipulation of artifacts and the physical world [21], and has been effectively applied to support children's learning in abstract science modeling [67], mathematics concepts [13, 33, 52], and AI literacy [1, 47, 49, 72, 75]. To enable easier knowledge transfer and generalization from learners' prior knowledge and concrete bodily experience to abstract AI literacy, we created a set of analogical and embodied representations that utilize learners' familiarity with bodily movements and bee pollination (Table 1). More details are introduced in the following sections.

Analogical Representations—Garden and Bee Analogies for the Mechanisms of Recommendation Systems. As an analogical representation, a *garden* is designed as an analogy for a recommendation system. *Flowers*, visualized by multiple features (e.g., color, petal size, height), represent items in the recommendation system. *Bees* pollinating flowers in the garden is the analogy for consumers of the recommendation system, and bee pollination represents user acceptance of a recommended item. An *environmental scientist* wants bees to visit and pollinate more diverse flowers and to maintain flower diversity. The environmental scientist's actions (Figure 3) demonstrate the mechanisms of a content-based recommendation system that considers data diversity to mitigate the filter bubble effect. A *businessman* is designed as an analogy for a regular AI recommendation system that causes filter bubbles.

Embodied Representation #1: Physical Distance to Represent Data Similarity/Distance in Multidimensional Feature Space. Existing research shows that higher-level bodily engagement leads to higher knowledge retention (i.e., not watching animations, or interacting with computers while sitting,

Table 1. The Connections between the Initial Analogical and Embodied Representations and the Target AI Concepts

	Representation	Target AI Concept
Analogical	Garden grows new flowers Bee and Bee pollination An environmental scientist or a businessman (1) studies which flowers the bee visited previously; (2) locates the flowers visited by the bees previously; (3) chooses the most different or similar flower within a certain range. Goal: Environmental scientist: Maintain the flower diversity to some extent. Businessman: Increase the flower similarity in the garden to make money	An AI recommendation system recommends new items (<i>AI application</i>) Users and users' acceptance of a recommended item (<i>AI application</i>) Inner working of content-based recommendation (<i>AI model</i>) (1) analyzes the items a user accepted before; (2) calculates the similarity between the item previously accepted by the user and other items; (3) recommends items similar enough to the previously accepted item, but still with a certain level of difference. Goal: Mitigate or cause the filter bubble effect (<i>Ethical issue</i>)
Embodied	Physical distance between two data objects: similar data objects are located close to each other Physical space for bodily movement: Being "trapped" by the smaller area with similar objects Physical blocks representing object features	Euclidean distance of multidimensional data (<i>Data</i>) Filter bubbles that decrease the information diversity in a recommendation system (<i>Ethical issue</i>) Multidimensional data described by multiple features (<i>Data</i>) and feature selection (<i>AI model</i>)

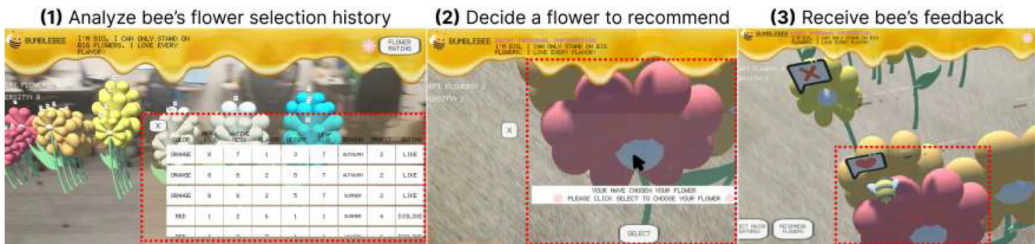


Fig. 3. Actions enacted by the environmental scientist: (1) analyze the bee's flower selection history; (2) locate the flowers similar to what the bee previously liked and recommend one; (3) receive a heart or a cross sign from the bee, indicating whether it accepts the recommended flower or not.

but involving bodily movements and locomotion) [37, 38]. We used physical distances in the real world to represent the Euclidean distances between multidimensional data objects. This embodied representation aims to support novice learners in using their full-body movements (e.g., walking, squatting down) and develop a more concrete and straightforward view of multidimensional data similarities/distances. For example, if a learner walks for a long distance from one flower to another, they understand that the Euclidean distance between the two multidimensional data objects that describe these two flowers is large. A learner squats down to view a shorter flower, deepening their understanding that the height feature of this data object has a lower value. In the implemented AR application, the flowers are virtual objects that support real-time rendering of flower locations based on their data similarities (Figure 4.1).

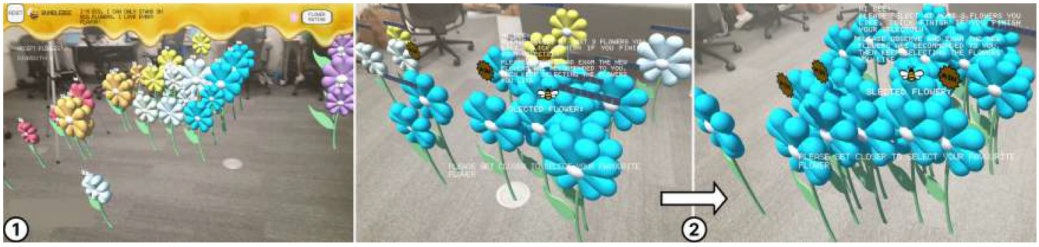


Fig. 4. (1) Data similarity visualized in physical space: The similar flowers are located closely, while dissimilar flowers are far away from each other. (2) Mimicking the filter bubble effect in a physical space: Learners are surrounded and trapped by similar flowers.

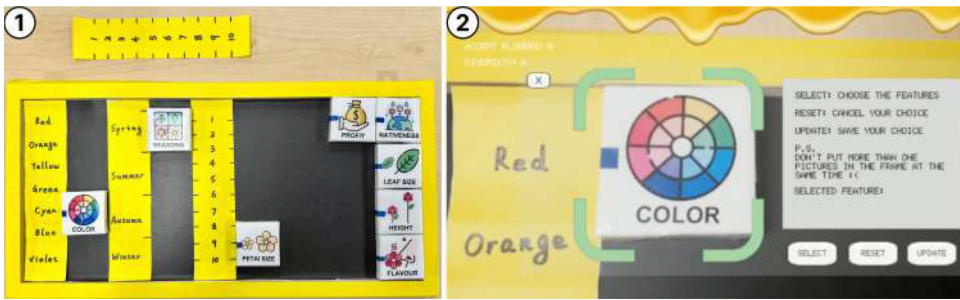


Fig. 5. Explore features with hand manipulation: Learners selected important features for data visualization and AI model building by picking up the corresponding feature block and scanning the image on it.

Embodied Representation #2: Full-Body Movements to Experience a Recommendation System and Filter Bubbles. Learners walk into the physical space of the garden and select flowers for pollination as a bee, which represents users accepting items in a recommendation system. With the learner selecting a flower, more similar flowers grow nearby. Such a natural phenomenon resembles the consumer's experience with a content-based recommendation system that recommends items similar to what the user has previously selected or paid attention to.

As learners visit more flowers based on a specific type of preference, more and more similar flowers will be growing clustered together in the garden (Figure 4.2). At the very beginning, learners can view and visit a very diverse range of flowers located in a large physical space. Gradually, learners no longer need to move around to select flowers because there are no more diverse flowers spreading out in the space. In the end, learners will be trapped/stationary in a much more limited space where flowers are still growing. This mimics the filter bubble effect formed in a recommendation system where consumers are no longer exposed to diverse information or options.

Embodied Representation #3: A Set of Physical Toolkits Supporting Hand Manipulation for Exploring Feature Selection. Feature selection is an important procedure for an AI model to learn the feature that matters the most. This embodied representation supports learners in selecting the important features of an AI model. Students play with a set of physical blocks (Figure 5), each of which represents individual data features from the multidimensional data describing flowers (i.e., flower color, leaf size, petal size, blossom season, height, flavor/scent, nativeness, and profit). By including different features in the model, the data representation and projection in physical space will be updated in real time.

Table 2. Basic Demographic Information of Teachers Who Participated in the Focus Group Interview

PID	Gender	Teaching Grade(s)	Teaching Subject(s)
s1p1	Female	8th	Earth Science
s1p2	Female	7th	Chemistry
s1p3	Female	7th	Biology/Life Sciences
s1p4	Female	8th	Chemistry, forensics

With these physical blocks, learners are supported to explore selecting different sets of features with direct hand manipulation. This design utilized the election and arrangement of physical objects, the blocks, to embody the abstract concept of “feature selection” to build the AI model by developing a symbolic understanding [11]. Such embodied interaction could potentially lower learners’ cognitive load.

5.3 The Implemented AR Learning Experience

By using the analogical and embodied representations above, we implemented two learning activities in a mobile-based AR application, including (1) role-playing a bee to experience a flower recommendation system and the filter bubble effect, and (2) role-playing an environmental scientist and a businessman to embody different mechanisms of the content-based recommendation system. Such AR-supported roleplaying aims to foster empathetic behaviors during perspective-taking by stimulating children’s imagination without disconnecting them from reality [68].

6 Study 2: Iterative Design and Evaluation with K-12 Students and Teachers

Building on the designs from Study 1 (see Section 3), we conducted iterative evaluations with students and teachers.

6.1 Research Method

6.1.1 Participants.

Students. We recruited 14 middle-school students through k-12 after-school programs and paired them into seven groups for the collaborative learning experience. Their ages varied from 10 to 14 (Mean = 12.3, SD = 1.1), comprising 8 boys, 5 girls, and 1 student who does not wish to identify his/her gender. Seven students received several hours of computer programming instruction at their school, while the remaining seven had no prior programming experience. Seven of them had little understanding of AI. Other participants had limited knowledge of AI as a smart robot, Google assistant, a technology that could harm humans, or as an advanced technology.

Teachers. Four K-12 science teachers also evaluated the design. They are all females and teaching grade levels ranging from grade 7th to grade 12th and a diverse range of STEM subjects (Table 2).

6.1.2 Study Procedure and Data Collection.

Evaluation with Students. Researchers set up the AR application on tablets. Two learners collaboratively used one tablet to go through the two learning activities in the AR application, each lasting for around 20 minutes. Two researchers facilitate the learning process upon request. Four Canon Camcorders were set up at each corner of the study areas, and one Canon Camcorder at the front of the desk where learners interact with the physical objects on the desk. We collected

Table 3. Mean (SD) of Pre- and Post-Tests on AI Concepts (N = 14)

Target Concept (Scores Range 0–3)	Mean_Pre (SD)	Mean_Post (SD)	Statistic	p value
Multidimensional data	1.00 (0.76)	2.57 (1.02)	0.00	0.003
Data similarity	0.79 (0.41)	0.93 (0.26)	2.50	0.371
Data distance	0.00 (0.00)	2.04 (1.01)	0.00	0.001
Filter bubble	0.86 (0.99)	2.32 (0.79)	0.00	0.003
Similarity-based model	0.00 (0.00)	2.86 (0.91)	−11.262	0.000
Feature selection	0.14 (0.35)	1.64 (0.58)	1.50	0.000
Recommendation algorithm	1.57 (1.24)	2.86 (0.77)	0.00	0.003
Filter bubble mitigation	0.57 (0.98)	2.64 (0.91)	2.00	0.000

Table 4. Participants' Attitude toward Different AI Concepts in Pre-Test and Post-Survey

Attitude Item (Question)	Multidimensional Data	Filter Bubble	Feature Selection	Recommendation System Mechanism
Relevance (e.g., Multidimensional data objects are relevant for my life)	1.214***	1.429***	0.5*	0.571
Curiosity (e.g., I am curious about discoveries in multidimensional data objects)	1***	1.143***	0.071	−0.5*
Enjoyment (e.g., I enjoy learning about multidimensional data objects)	0.786***	1.214***	0.786*	0.643
Self-efficacy (e.g., I believe I can improve in learning multidimensional data objects)	0.429	0.429	0.286	0.286

*p < 0.05; **p < 0.01; ***p < 0.001.

data through pre- and post-tests on target AI concepts, interviews, screen recordings, and video recordings of the system interactions from 14 participants.

Evaluation with Teachers. With researchers' facilitation, teachers experienced the AR application while doing "think-aloud," where they vocalized their thoughts and reactions about the two learning activities. This activity aimed to provide teachers with first-hand experience of the application and to capture their real-time feedback during their interaction with the system. A focus group interview with teachers was conducted to collect additional feedback and discuss design solutions to address the challenges identified from student evaluation.

6.1.3 Measures and Data Analysis. To evaluate how the design outcome affects children's AI literacy, we measured students' AI knowledge and attitudes before and after interacting with the system by pre- and post-surveys (Tables 3 and 4).

Student Learning Gains. Under each AI literacy category, we selected specific AI concepts related to the AI recommendation system by following the design guidelines of the "Big Idea #2 Representation and Reasoning" and "Big Idea #3 Learning" developed by AI4K12 [65] and the specific AI literacy

competency summarized by the existing literature review [48] on what AI literacy should be considered for K-12 education. A coding scheme was developed to analyze the open-ended answers to knowledge assessment questions. Two researchers coded all the questions independently and achieved a Cohen's kappa of at least 0.74 for each question, indicating substantial agreement. The relative learning gains were measured by evaluating the difference between pre- and post-tests.

Student Attitude Shifts Related to AI. Participant attitudes were evaluated by their intrinsic motivation and self-efficacy in pre- and post-surveys (Table 4, Attitude item). The questions are adapted from an existing motivation survey assessing students' motivation for learning CS/technology with AR [55], which is an adapted version of the Science Motivation Questionnaire II for the usage in technology or computer science education. The attitude questions assess students in two aspects, namely motivation and self-efficacy.

Student Learning Behaviors and Feedback. We also analyzed students' learning behaviors and feedback on the experiences by going through camera and screen recordings. Two researchers independently analyzed learners' verbal and non-verbal behaviors during the three learning activities and then discussed the themes generated for each pair of participants. Disagreements are resolved iteratively during the discussion. Seven pairs of participants are categorized into a high-performance group and a low-performance group based on their learning performance during the learning activities. The high-performance group contains pairs of participants who finished learning activities in a shorter time and achieved higher accuracy. Low-performance groups finished the activities in a longer time with lower accuracy.

Focus Group Interview with Teachers. Based on the learning challenges identified from the proof-of-concept study with middle-school students, we designed specific questions for the focus group interview with the four K-12 teachers. Two researchers conducted a thematic analysis and extracted design insights from teachers' feedback.

6.2 Findings

6.2.1 RQ2: Learning Challenges and Affordances of the AR Prototype.

AI Learning Gains. The data for all assessment questions except Q5 was not normally distributed based on the Shapiro-Wilk test. Thus, we applied the paired t -test on the pre- and post-test scores for Q5 and a non-parametric test, the Wilcoxon signed-rank test, for the rest of the questions. Significant statistical differences are identified in the pre-post test for all the AI knowledge components except for data similarity, indicating that the design elements and the learning activities effectively improved learners' understanding of the four categories of AI literacy (Table 3). For data similarity, 80% of the learners got full scores on the pre-test. This may indicate that this question is too easy to demonstrate a significant increase in the specific AI knowledge component. Although the rest of the differences are statistically significant, the small sample size limits the generalizability of the findings. These results should be considered exploratory and in need of replication with larger samples.

AI-Related Attitude Shifts. We compared participants' attitudes toward different AI concepts before and after each learning activity (Table 4). After experiencing the design elements for learning multidimensional data and the filter bubble, there is a significant positive attitude shift for Relevance, Curiosity, and Enjoyment. For feature selection, there is a significant positive change in Relevance and Enjoyment. Using embodied metaphors and analogies implemented in AR helps to bridge the physical world and virtual information and also makes learning more enjoyable for learners. For the

recommendation system, there is a significant negative shift in curiosity. This might be caused by the demystifying of the basic working mechanism behind content-based recommendation systems.

Attitude Shift with Embodied Metaphors. Nine out of the 14 participants explicitly mentioned that they found the walking experience highly enjoyable and were more motivated to learn with such a walking experience in the AR garden. Different reasons were reported. First, four students enjoyed walking inside the flowers to learn: “It’s like walking in a real garden. I really enjoyed being part of the experience.” Two of them also mentioned that they preferred walking in a garden with more colorful and diverse flowers. This naturally led to their reflection on filter bubbles and potential design considerations for a more ethical recommendation system. Second, compared to 2D interfaces, six students found it a fun learning experience to walk around and have full-body interactions for learning: “I like moving around and first-hand experience. It’s more interesting than in classrooms.” Third, four students pointed out that walking for different distances for flower observation, along with hands-on manipulation for feature selection, makes multidimensional data and the AI model much more concrete and intuitive to understand.

Attitude Shift with Analogies and Role-Play. By analyzing learners’ feedback on their learning experiences with analogies and role-play, we identified a relatively mixed attitude shift. Although the self-reported curiosity for how recommendation systems work decreased, students reported that role-playing analogies of AI concepts supported them in developing a deeper reflection on the working mechanism behind recommendation systems. For example, one participant started wondering why the flowers that were not preferred by the bee were still there in the garden. Some participants thought more about the end goals of the recommendation system designer: “If the businessman only gives bees similar flowers, will bees be unhealthy? So the businessman doesn’t care if the bees are unhealthy?” and the information gap in end-users: “Do the bees know that eating similar flowers will be unhealthy?”. All the participants transferred what they learned from the analogies and role-play and developed some initial thoughts on how to design a recommendation system to mitigate the filter bubble effect. In addition, by role-playing recommendation systems, one participant reported their appreciation of the intelligence of recommendation systems: “The recommendation system is really smart because it can guess what the bees like.”

Learning Behaviors. Learning with Embodied Metaphors. By pairing learners’ non-verbal behaviors with their learning performance during learning activities, we identified specific body movements that lead to higher performance in learning target AI literacy (see Figure 6(c)–(e) as examples).

For learning *data similarity*, we observed that all four high-performance groups utilized body movements to pay extra attention to data attributes of a flower that are not very visually obvious. For example, group 5 observed the virtual flowers along with their floating data vectors from both the front side and back side to evaluate if they ignored any data attributes. Learners also provided feedback confirming that the walking experience that uses physical distance to embody data similarity enabled their self-discovery of the corresponding data literacy.

While learning *feature selection*, three out of four high-performance groups utilized different body movements collaboratively to achieve better AI model performance. For example, one learner from group 3 went back to the physical object table and updated the predicted feature value through hand manipulation, while the other learner kept exploring the flower recommendation options by walking in the AR garden.

During role-playing *the content-based recommendation system*, three out of four high-performance groups pointed to virtual flowers in the physical space rather than pointing to the flower on the tablet screen. This indicates that the body movement of item selection in a physical space instead of on a 2D interface may better embody the underlying abstract AI concept, and thus, better facilitate learners’ understanding. Furthermore, we also observed that one high-performance group made

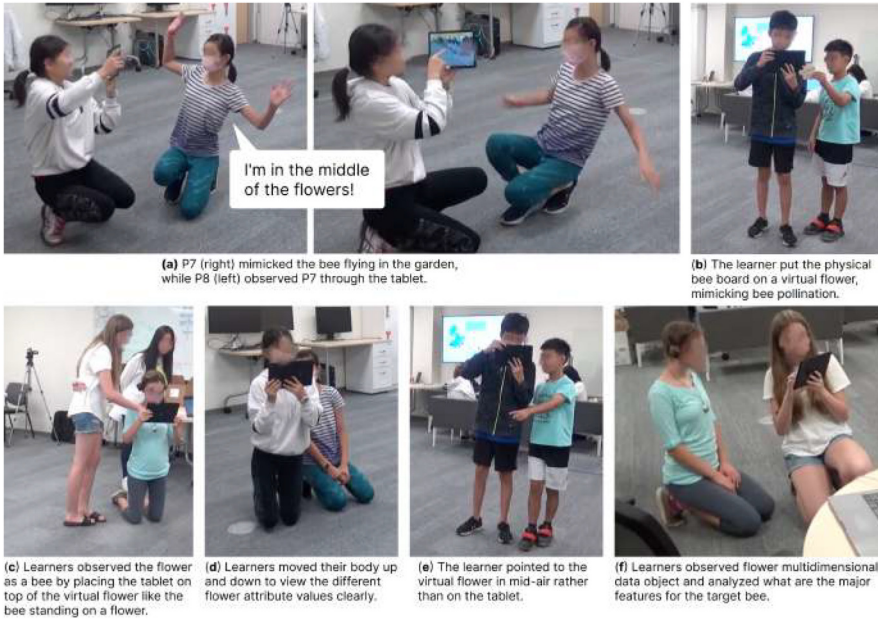


Fig. 6. Middle-school students interacting with the embodied metaphors and analogies implemented in BeeTrap.

use of walking to intentionally recommend more diverse flowers, different from their previous recommendations. While walking in the AR garden, when learners from group 2 saw a flower that had been selected by the bee, they turned to another direction in the room to find more diverse flowers.

While experiencing *filter bubbles* formed by the flower recommendation system underlying the AR garden, we found that high-performance groups tended to be more physically active than the low-performance groups. For example, group 3 explored and observed the changes in the garden after each round of flower selection by walking around more actively, while group 4 mostly stood still and only rotated the tablet to observe the changes in flowers.

Learning with Analogies and Role-Play. To understand how learners' learning performance is influenced by role-playing analogies of AI concepts, we focused on analyzing the verbal and nonverbal behaviors during learners' roleplay of content-based recommendation systems and the bee (see Figure 6(a) and (b) as examples).

For learning performance in *role-playing recommendation systems*, we found all groups successfully simulated the high-level content-based filtering process for a recommendation system. With all the different data sources at hand, they first analyzed the bee's profile information to formulate an initial hypothesis on the essential flower features that determine the bees' flower preferences. Then, they switched between the analysis of the bees' flower selection history and the bee's like or dislike response to their latest flower recommendation. This indicates the effectiveness of role-playing the analogies of AI recommendation systems in developing young learners' AI literacy.

From the *role-playing bee* learning activity, we observed the unique learning benefit that role-playing brought to the learning of the filter bubble. All four high-performance groups demonstrated a higher level in role-playing. For example, group 1 and group 3 placed the physical board of the bee to mimic the bee pollination process. Group 3 constantly used verbal behaviors to reinforce their role-play, such as "Just bee-lieve it" or "I'm flying through the garden." They also discussed

what environmental factors may affect the production of new flowers in a real garden. Group 2 also made the connection between their flower pollination during role-playing the bee and the changes in the garden where more flowers that are similar to their selection were generated.

We also observed that role-playing analogies of AI concepts effectively helped learners shift their views and develop a basic understanding of both the technical and the ethical aspects of AI recommendation systems. For example, when being asked how to avoid filter bubbles in daily life, P8 mentioned from a user's perspective that we should watch different things, and also from a recommendation system, that it should recommend different suggestions on what the users like.

6.2.2 RQ3: Insights for Design Iteration. First, we need analogical and embodied elements explicitly to represent major steps in the diversification algorithm. Middle-school students had lower learning gains on filter bubble mitigation. After looking into learners' feedback after the learning experience, we found that learners spent most of their efforts playing with the physical blocks representing data features and making intelligent guesses on what flowers to recommend. However, the actual mitigation strategy, the diversification algorithm, remained vague. Teachers suggested adding more concrete representations to enhance learners' interaction and thus deepen their understanding of the algorithmic steps involved in a recommendation system. Relying solely on an analogical representation of the environmental scientist aiming to diversify flowers in the garden may not be sufficient.

Second, the user history table imposed unnecessary cognitive demands for calculating multi-dimensional similarity computation, which was not central to the learning goals. We observed two learning behavior patterns. One type of learner only focused on comparing data points in the user history table without fully experiencing the embodied learning in the garden. This indicates that the segmentation between the data table and the physical space and garden analogies may distract students' attention from the target AI concepts. The other type of learner mainly relied on a trial-and-error strategy by making intelligent guesses on what flowers to recommend without adequately analyzing the numeric table to identify the shared features among flowers liked by the target bee. This indicates the insufficient learning support provided by the data table. Students provided feedback that the numeric values in the user history table were too overwhelming for them. In the focus group interview, teachers suggested creating a more integrated analogy in the garden context to replace the numeric table for user selection history. They found the initial design to be both overwhelming and disconnected from other representations in the garden. Teachers reported that the constant attention switch between the data table and the garden distributed in the physical space added unnecessary cognitive loads to the problem-solving process.

Third, a better data representation is needed to form a more accurate understanding of the multidimensional feature space. The pre- and post-tests showed an insignificant increase in students' understanding of data similarity. After looking into students' misconceptions about data similarity, we found that learners focused too much on the petal color feature, a visual feature that tended to dominate their attention.

Fourth, we noticed that some students demonstrated a positive attitude toward the filter bubble during the interview, as they liked the experience of only being recommended the specific type of flowers they liked. This is aligned with emerging findings on children's perception of filter bubbles. This calls for a new design for children to learn more about the potential impact of filter bubbles in terms of the loss of information diversity and the health of the information diet.

Last, from the instructional design perspective, teachers proposed to add an independent learning activity, alongside the role-playing of the environmental scientist, to help students understand the mechanisms of a recommendation system that does not consider diversification.

7 Discussion

We discuss insights from the iterative studies introduced in Section 3 and elaborated in Sections 4 and 6.

7.1 How Embodied Metaphors and Role-Playing Analogies Support Learning AI Literacy

By analyzing learners' verbal and nonverbal behaviors during three learning activities, we observed patterns of how learners utilized body movements, including walking, gestures, and hand manipulation, to learn abstract AI concepts. First, learners utilized their horizontal high-level body movements in the physical space to find the flowers that have certain values for certain data attributes. They also constantly move their body up and down to observe specific values of flowers, which further deepens their understanding of different attributes and their different values in a multidimensional data point. This indicates the effectiveness of using embodied metaphors, including UP-DOWN and LEFT-RIGHT, for value decreases and increases in multidimensional feature space [7]. We also observed the verbal and non-verbal behaviors of learners exploring data visualization with different data attributes by hand manipulation of physical blocks representing individual data attributes. This is aligned with existing theories of dynamic visualization and embodied learning in: (1) making or observing gestures, (2) manipulating and interacting with objects, (3) using body metaphors, and (4) using eye movements as retrieval cues [19].

After evaluating the learning effectiveness of interacting with analogies, we found that walking through the virtual flowers located in the physical space supported learners to successfully take the perspective of the bee in the second learning activity and the recommendation system in the third learning activity. Such role-playing enhanced learners' understanding by connecting their prior knowledge of flower pollination with the working mechanism behind content-based recommendation systems. Being physically surrounded by overwhelming and similar flowers during role-playing bees also emphasized learners' experience of filter bubbles and, in turn, deepened their AI literacy learning. Such findings further validate the learning benefit of perspective-changing that is uniquely supported by AR. AR-supported role-play provides more empathic behaviors during perspective-taking because AR excites children's imagination without forcing them to lose contact with reality [68].

7.2 Interaction between Embodied Metaphors and Analogies

Our findings also indicate the synergistic relationship between embodied metaphors and analogies. First, we observed learners' body movements that support their role-playing. For example, group 1 mimicked the bee's pollination behavior during the bee role-playing activity by walking closer to certain flowers, using the physical object representing the bee to hover on the flower as an action for selection. Second, we noticed the positive impact of role-play on triggering more embodied experiences for learners to embody abstract AI concepts. For example, during role-playing analogies of recommendation systems, learners naturally pointed to the virtual flowers in mid-air as a gesture of selection. Such body movement makes the concept of "item recommendation" a more concrete action. Similarly, the role-play of a bee makes learners physically surrounded by similar flowers, which embodies the abstract concept of being surrounded by similar information in the filter bubble. Such interactions between body movement and analogies connect with the existing theory that the bodily experiences of learning targets occur when bodily movements are semantically related to these targets [19, 24, 34].

7.3 Cognitive Load Mitigation Strategies

Our findings indicate a few design strategies to reduce cognitive loads introduced by complex AR elements with underlying AI concepts.

First, integrate representations to avoid split-attention. Instead of having a separate data table, data representation can be more directly embedded into the AR garden (e.g., feature values appear as in-scene tooltips rather than in a separate pane). This could reduce the need for learners to split attention between formats, mitigating the split-attention effect described in Cognitive Load Theory [63]. Furthermore, by applying Mayer's Coherence Principle to replace text tables with color-coded overlays, learners can process only germane information and avoid extraneous load [54].

Second, chunk and scaffold complexity. For example, introduce only two attributes at first (e.g., color + size), then gradually add more, and precede open exploration with a worked example. This leverages scaffolding to manage intrinsic load and support schema acquisition [69, 71].

Third, use just-in-time prompts and guidance. In the next iteration, context-sensitive nudges ("Recall: focus on spatial proximity now!") and step-by-step progress indicators could be provided. This aligns with principles of guided instruction and adaptive feedback shown to support novice learners' cognitive processing [32, 70].

8 Limitations and Future Work

8.1 Limitations

Limited Sample Size and Scope. The first study only involved seven AI experts, and the evaluation also involved a relatively small number of participants (14 middle-school students and 4 STEM educators), which may restrict the generalizability of the findings across diverse educational settings and age groups.

Potential Oversimplification of Complex Concepts. While the embodied and analogical representations help make abstract AI ideas more accessible, there is a risk of oversimplifying key algorithmic details. For instance, the environmental scientist character and the tangible representations may not fully capture the nuances of mitigation algorithms for filter bubbles, potentially leading to misconceptions.

Transferability of Design Elements. The design choices, particularly the use of metaphors like bees, flowers, and gardens in an AR setting, may be effective in this specific context but could face challenges when adapted to other AI learning objectives or when targeting different age groups. Without developing a design case in another analogy context for a different set of AI concepts, the effectiveness of such an iterative multi-phase design pipeline is still preliminary and requires further validation.

8.2 Future Work

We envision such a set of embodied and analogical representations of AI concepts to have a low floor (i.e., easy for children to learn and use), a high ceiling (i.e., able to advance children in making more complex projects), and wide walls (i.e., inclusive for learners with different interests). For a high ceiling, data visualization in the 3D space can be further designed to introduce more advanced mathematical concepts in multidimensional and multimodal data; the AR garden mimicking the filter bubble can be extended to different goals of different stakeholders of the recommendation system, data privacy and digital privacy issues [7], bias issue, transparency, trust [19], and so on; the physical toolkits for exploring feature selection should be replaced with more contextualized embodied elements based on the empirical findings; the flower selection behavior during the role-play can be expanded to the hidden feature engineering in deep learning, collaborative filtering recommendation

systems, overfitting, and so on. To achieve wide walls, more refined body movements and physical space exploration can be introduced to stimulate young learners' interest. The recommendation system can be contextualized in more diverse scenarios, such as recommending music, movies, restaurants, snacks, or books based on learners' backgrounds and preferences.

9 Conclusion

In conclusion, this work demonstrates an iterative multi-phase design approach that designs and integrates embodied metaphors and analogies to effectively enhance AI literacy among middle-school students. Through a collaborative process involving AI experts, K-12 educators, and students, we developed a design case, a tablet-based AR application that translates abstract AI concepts—such as recommendation systems and filter bubble effects—into tangible, relatable experiences. Our mixed-method evaluations indicate significant improvements in students' understanding of these concepts, although they also reveal challenges such as cognitive overload, an overemphasis on visually dominant features, and difficulties in conveying nuanced algorithmic details. These findings specified the design iterations needed and yielded actionable design implications. This iterative design process illustrates how various types of design insights are elicited and collected at different stages with distinct stakeholders. This emphasizes the importance of continuous evaluation and redesign in developing educational tools for complex subjects like AI. Future work will explore scaling these design principles and addressing the identified challenges to foster a deeper, more comprehensive understanding of AI among young learners.

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